CURRICULUM VITAE

Huiyi (Cheryl) Wang *March 2026*

PERSONAL INFORMATION

Phone: (438)-522-0430
E-mail: huiyi.wang@mail.mcgill.ca/cheryl.wang.huiyi@gmail.com
Personal Webpage: https://cherylwang20.github.io
GitHub: github.com/cherylwang20

DEGREES

McGill University, Faculty of Engineering *Sep 2023 - Present*
Doctor of Philosophy in Mechanical Engineering with focus on Robotics Control with State of Art Machine Learning (ML) Technique

McGill University, Trottier Space Institute at McGill (TSI) *Sep 2021 – Aug 2023*
MSc Physics with Thesis GPA: 4.00/4.00
Thesis: Constraining the Formation Scenario of Earth through Imprints of Primordial Solar Nebula within the Deep Mantle.

University of California, Los Angeles (UCLA) *Sep 2017 – Mar 2021*
Bachelor of Science in Astrophysics, with Honors GPA: 3.70/4.00
Honor Thesis: Gravitational-Wave Signatures from Compact Object Binaries in the Galactic Center

TECHNICAL SKILLS

Coding Language: Python, C++, C#, Java (basic), MATLAB, PyTorch
Skills: Reinforcement Learning, Dynamic Programming, Sequential Decision Making, Deep Learning, Generative AI, Robotics, Muscle Driven Simulations, Musculoskeletal Modeling, Computer Visions
Certificate: Amazon Web Service (AWS) Certified Machine Learning – Specialty (Certificate)

RESEARCH & PROFESSIONAL EXPERIENCE

NVIDIA – Generative AI Engineering *Mar 2026 – July 2026*
Internship, Shanghai, China

- Project details are currently confidential and cannot be shared at this time due to an active Non-Disclosure Agreement (NDA).

EPFL – Scaling up Fullbody Musculoskeletal Simulation *Sep 2025 - Present*
Ph.D. Researcher Exchange, Hybrid, Geneva, Switzerland

- Created the MyoBimanualArm and MyoFullbody model based on MyoSuite model, with full contact and symmetry enabled, with muscle path jumping corrected.
- Refine the AMASS dataset to MyoFullBody retargeting pipeline via GMR + SMPL fitting, and improve the joint violations by up to 12% and RMSE by 10cm.
- Scale up the reinforcement learning and imitation learning with GPU acceleration via MuJoCo Warp to achieve large scale fullbody motion. Allowing billions of training steps to be completed within 48 hours, which previous takes weeks.

McGill / MyoSuite - Developing Physiological Accurate Controller for Elderly *Sep 2023 - Present*
Ph.D. Researcher & Developer, MyoSuite, Hybrid, Montreal, Canada

- Developed a 210-muscle actuated MyoBack model in MyoSuite, integrating with current lower limb model (80 actuators) to achieve a full body neuromusculoskeletal model that is controllable.
- Implemented Proximal Policy Optimization (PPO) with MyoSuite musculoskeletal model to replicate aging conditions such as sensorimotor delay and sarcopenia in virtual agent balancing task.
- Designed a Hierarchical RL framework to model human decision-making in balance control, improving humanoid robots' adaptability and stability in dynamic environments.

Reinforcement Learning with UR10e Robotic Arms

Internship, National Research Council Canada, Ottawa *May 2024 - June 2025*

- Created the first Gymnasium environment integrating the UR10e robotic arm for vision-based reinforcement learning capable of benchmarking tasks such as pick and place.
- Led the integration of online training for novel mask-based goal-conditioning for robotic manipulation using JAX and stablebaseline3, resulting in a 15% improvement in the adaptive accuracy of reaching tasks in sim2real transfer.
- Developed advanced motion planning for the UR10e robotic arm using computer vision and pre-trained object segmentation model to increase reaching precision and speed by ~20% and allow zero-shot generalizations.

PUBLICATION

1. **H. Wang,** J. Kovecses and G. Durandau, (2025) "*Reinforcement Learning Identifies Age-Related Balance Strategy Shifts*," in IEEE Transactions on Neural Systems and Rehabilitation Engineering
2. **Wang, H.**, Tan, C., et al., 2025, *"MyoChallenge 2024: A New Benchmark for Physiological Dexterity and Agility in Bionic Humans",* NeurIPS 2025 Datasets and Benchmarks Track
3. **Wang, H.,** Stephan, A. P., Naoz, S., et al. 2021. *"Gravitational-Wave Signatures from Compact Object Binaries in the Galactic Center."* https://iopscience.iop.org/article/10.3847/1538-4357/ac088d
4. **HUIYI WANG**, Fahim Shahriar, et al. "*Versatile and Generalizable Manipulation via Goal-Conditioned Reinforcement Learning with Grounded Object Detection*", CoRL 2024 Workshop on Mastering Robot Manipulation in a World of Abundant Data
5. Chun Kwang Tan, **Cheryl Wang**, et al. "*MyoAssist 0.1: MyoSuite for Dexterity and Agility in Bionic Humans*", ICORR 2025 Conference Paper
6. Rohan Walia, Morgane Billot, Kevin Garzon-Aguirre, Swathika Subramanian, **Huiyi Wang**, et al. "*MyoBack: A Musculoskeletal Model of the Human Back with Integrated Exoskeleton*", ICORR 2025 Conference Paper
7. Vittorio Caggiano, Guillaume Durandau, **HUIYI WANG**, et al. *"MyoChallenge 2023: Towards Human-Level Dexterity and Agility"*, Proceedings of Machine Learning Research (PMLR)
8. Vittorio Caggiano, Guillaume Durandau, Seungmoon Song, Chun Kwang Tan, **Huiyi Wang** et al., *"MyoChallenge 2024: Physiological Dexterity and Agility in Bionic Humans"*, NeurIPS 2024 Competition Track. https://openreview.net/forum?id=2vkf3Wkw2v

TALKS AND PRESENTATION

1. **"Getting started with MyoSuite, Reinforcement Learning"**, McGill University, *MECH 561. Biomechanics of Musculoskeletal System*s, invited seminar, Feb 3rd, 2026
2. "**Simplifying Goal Representations with Masks in Vision-basedReinforcement Learning",** National Research Council Canada (NRC), *DTKX Seminar Series Talk*, Aug 13th, 2025.
3. **"Machine Learning Driven on Full Body Model Musculoskeletal Simulation",** Neuroprosthesis Symposium, *Invited Student Talk*, June 12th, 2025.
4. **"A Reinforcement Learning Approach for Assessing Standing Balance in Sarcopenia"***, Canadian Society of Biomechanics (CSB), *Oral Presentation*, 20th Aug 2024.
5. **"MyoSuite/MyoChallenge: Towards Full-Scale Human Embodied Intelligence",** *Guest Lecture,* The Theoretical & Computational Neuroscience Journal Club, Johns Hopkins University, 11th Sep 2024
6. **"Ariel Mission Target: A New Design for Phase Curve Observation"** Trottier Space Institute, McGill University Cowan Group Conference, 9th Aug 2023
7. **"Constraining the Earth Formation through Solar Nebular Imprints with Deep Mantle,"** Trottier Space Institute, Lee Research Group Meeting, 7th Dec 2022

8. **“CONSTRAINING THE HUBBLE CONSTANT: From the Tension Between PLANCK (CMB) and SHoES (SnIa) to the Cutting-Edge GW and FRB Detections,”** *Poster Presentation*, Trottier Space Institute, Bell Room, 6th Dec 2021
9. **“Gravitational Wave Signatures from Compact Object Binaries in the Galactic Center,”*** 52nd Annual DDA Meeting Virtual, 18th May 2021.

(*) Accepted with a peer reviewed abstract.

WORKSHOP ORGANIZATION

MyoConference 2026 | Organizer & Session Lead
Hybrid |Feb 15th 2026

- Directing the planning and execution of the Biomechanical Session of MyoConference online. With more than 70 participants around the globe to encourage open-source collaboration with MyoSuite.

NeurIPS 2025: MyoSymposium 25 | Organizer & Performance Baseline Provider
San Diego, CA |December 2025

- Managed a global competition reaching over 15 countries and receiving 500+ submissions, while coordinating an advocacy team of 20+ individuals.
- Directed the manipulation track to build a customized reinforcement learning environment featuring a musculoskeletal simulation agent, successfully open-sourced at MyoHub.
- Streamlined the full software development lifecycle from architectural design to PyPI deployment, fostering broad adoption in robotics and biomechanical research.

NeurIPS 2024: MyoSymposium 2024 | Organizer & Performance Baseline Provider
Vancouver, Canada |December 2024

- Engineered and refined advanced environments for the real-time control of musculoskeletal agents.
- Maintained production-level software standards to support rapid prototyping and testing of advanced reinforcement learning algorithms by the wider research community.

NeurIPS 2023: MyoSymposium 2023 | Organizer & Advocacy
New Orleans, LA | December 2023

- Collaborated in the organization of the MyoChallenge at NeurIPS. Helped with public relations, workshop organization, and communication with participants.

AWARDS AND HONORS

1. **MEDA (McGill Engineering Doctoral Award)** - McGill Faculty of Engineering *2023 - 2025*
2. **Best Student Presentation** – Neuroprosthesis Symposium *2025*
3. **Best Pitch Presentation –** Wearables and Biomechanics Pitch Competition *2025*
4. **Graduate Mobility Award** – McGill Faculty of Engineering *2025*
5. **Catherine Berg Student Travel Award** – CRIR-HJR *2025*
6. **GREAT Travel Reward** - McGill Faculty of Engineering *2024, 2025*
7. **Louis Ho Fellowships** - McGill Faculty of Engineering *2025 – 2026*
8. **Thomas and Penelope Deirdre Szirtes Fellowships in Engineering** - McGill *2024 – 2025*
9. **Geoff Hyland Fellowship -** McGill Faculty of Engineering *2023 - 2024*
10. **Grad Excellence Award**– McGill Physics/Engineering Department *2021, 2022, 2023, 2025*
11. **Trottier MSI Graduate Award** - McCall MacBain Scholarships, McGill University *2021 – 2023*
12. **College Honors Program** - UCLA College Honors *Spring 2020 – June 2021*
13. **UCLA Departmental Highest Honor** – UCLA Department of Physics and Astronomy *June 2021*
14. **UCLA Dean’s Honor List** – UCLA College of Letter and Science *Fall 2017, Winter 2019, Winter 2020, Spring 2020, Fall 2020, Winter 2021*

JOBS AND TEACHING ROLES

1. **Teaching Assistance, Applied Electronics and Instrumentation – MECH 383**
   McGill, Faulty of Engineering *Jan 2025 – Apr 2025*
2. **Teaching Assistance, Mechanics II - MECH 220**
   McGill University, Faculty of Engineering *Jan 2024 - Dec 2024*
3. **ANS Platform Specialist**
   McGill University, Montreal *July 2022 - Apr 2024*
4. **Graduate Course Assistant – Classical Mechanics, PHYS – 101/131**
   McGill University, Montreal *Sep 2022 – Dec 2022*
5. **Teaching Assistant, Intro Physics – Electromagnetism, PHYS – 102**
   McGill University, Montreal *Jan 2022 – April 2022*
6. **Teaching Assistant, Climate Physics, ATOC/PHYS - 404**
   McGill University, Montreal *Sep 2021 – Dec 2021*

OUTREACH

1. **MyoChallenge Advocacy** - MyoSuite *June 2024 - Present*
   Description: Led a 40-member advocacy group to organize workshops, tutorials, and talks for MyoChallenge 24' (competition hosted at NeurIPS) participants.
2. **Pen Pal Program -** Letters to a Pre-Scientist *Sep 2023 - Aug 2024*
   Description: Engaged with minority students in suburban areas in the US, fostering their interest in STEM professions by exchanging letters that introduced them to my own profession and experiences.
3. **AstroMcGill Space Fair Volunteer -** McGill Outreach *May 2023*
   Description: Presented explanations on space and ground telescope design to primary students during Science Day and guided them in the hands-on construction of their own telescopes using LEGO.
4. **Physics Matter Society** – Volunteer at Trottier Space Institute *Oct 2021 - Aug 2023*
   Description: Contributed to an outreach organization's mission to popularize and demystify physics for the general public.
5. **Space Explorer** – Volunteer to teach at Montreal Elementary Schools *Oct 2021 – Oct 2022*
   Description: Designed and implemented physics course modules and labs tailored to elementary students, fostering their interest in science.